LUNARSAFE // MATHEMATICAL SPECIFICATION & ALGORITHM SUITE

RIGOROUS DERIVATIONS, ERROR PROPAGATION & CRYPTOGRAPHIC SEALS

STANDARD: Horn (1981) Formulation · LOLA Error Propagation · ISRO 24m Standard

SECTION 1 — HORN (1981) DEM SLOPE ALGORITHM: FULL DERIVATION

In digital terrain analysis, local surface normal vectors and physical terrain gradients must be computed from discrete elevation matrices. LunarSafe implements the third-order finite difference method established by Horn (1981). Consider a 3x3 window of elevation nodes centered at pixel (x, y) with horizontal ground sample distance GSD (meters/pixel):

3x3 LOCAL ELEVATION MATRIX STENCIL

```
[ z00   z10   z20 ]
[ z01   z11   z21 ]
[ z02   z12   z22 ]
```

The orthogonal surface gradients p = dz/dx and q = dz/dy are computed using Horn's distance-weighted stencil with 8*GSD normalization:

HORN (1981) GRADIENT FORMULAS & SLOPE ARCTANGENT

```
p = dz/dx = [ (z20 + 2*z21 + z22) - (z00 + 2*z01 + z02) ] / (8 * GSD)
q = dz/dy = [ (z02 + 2*z12 + z22) - (z00 + 2*z10 + z20) ] / (8 * GSD)

Slope (radians):  S_rad = arctan( sqrt( p^2 + q^2 ) )
Slope (degrees):  S_deg = S_rad * (180.0 / PI)
```

Citation: Horn, B.K.P. (1981). "Hill shading and the reflectance map." Proceedings of the IEEE, 69(1), 14-47. doi:10.1109/PROC.1981.11918. Horn's formulation provides an optimal compromise between high-frequency noise attenuation and spatial fidelity, offering zero phase distortion on planar terrain.

SECTION 2 — SOBEL GRADIENT EDGE DETECTOR & OPTICAL CALIBRATION

When evaluating 2D single-band optical imagery (where direct DEM elevations are unavailable), slope gradients are approximated via convolution with Sobel kernels Kx and Ky. For 8-bit unsigned integer data [0, 255], the theoretical maximum Sobel gradient magnitude is:

THEORETICAL MAXIMUM SOBEL & CALIBRATED OPTICAL SLOPE

```
Gx_max = 4 * 255 = 1020,   Gy_max = 4 * 255 = 1020
 G _ m a x     =   s q r t (   1 0 2 0 ^ 2   +   1 0 2 0 ^ 2   )   =   1 0 2 0   *   s q r t ( 2 )  "H   1 4 4 2 . 4 9 7 8
Ratio  = min( 1.0,  sqrt( Gx^2 + Gy^2 ) / 1442.4978 )
Slope  = arcsin( Ratio ) * (180.0 / PI)   [Clamped to 0.0° - 90.0°]
```

SECTION 3 — ISRO 24m GRID CLASSIFIER: DISCRETIZATION

To strictly enforce ISRO's Chandrayaan landing box standard without spatial distortion, grid cell partitioning uses integer flooring (Math.floor) rather than nearest-integer rounding (Math.round). This guarantees that every raster cell contains an integer number of pixels:

GRID CELL SAMPLING & AREA EQUATIONS

```
pixelsPerGrid    = Math.floor( gridSizeMeters / gsdMetersPerPixel )
actualCellMeters = pixelsPerGrid * gsdMetersPerPixel
cellAreaM2       = actualCellMeters * actualCellMeters
gridCols         = Math.floor( width / pixelsPerGrid )
gridRows         = Math.floor( height / pixelsPerGrid )
```

SECTION 4 — FIRST-ORDER ANALYTICAL SLOPE UNCERTAINTY PROPAGATION

Let elevation errors be independent and identically distributed (i.i.d.) Gaussian random variables with variance sigma_z^2. Propagating this variance through Horn's finite difference gradients:

GRADIENT VARIANCES & COVARIANCE DERIVATION

```
Var(p) = sum_i (dp/dz_i)^2 * sigma_z^2 = [ 1^2 + 2^2 + 1^2 + 1^2 + 2^2 + 1^2 ] / (64 * GSD^2) * sigma_z^2
Var(p) = 12 / (64 * GSD^2) * sigma_z^2 = 3 / (16 * GSD^2) * sigma_z^2
Var(q) = 3 / (16 * GSD^2) * sigma_z^2
Cov(p, q) = 0   [Symmetric stencil cancellation renders p and q uncorrelated]
```

Propagating into the slope S = arctan(r) where r = sqrt(p^2 + q^2) = tan(S):

SLOPE UNCERTAINTY EQUATION (CLOSED-FORM)

```
(dS/dp)^2 + (dS/dq)^2 = 1 / (1 + r^2)^2 = 1 / (1 + tan^2(S))^2 = cos^4(S)
sigma_S^2 = Var(p) * [ (dS/dp)^2 + (dS/dq)^2 ] = [ 3 * sigma_z^2 / (16 * GSD^2) ] * cos^4(S)

sigma_S_rad = [ sqrt(3) * sigma_z / (4 * GSD) ] * cos^2(S)
sigma_S_deg = [ sqrt(3) * sigma_z / (4 * GSD) ] * cos^2(S) * (180.0 / PI)
```

SECTION 5 — CONSERVATIVE 2-SIGMA HAZARD CLASSIFIER

Under the conservative safety rule, a candidate landing cell is classified as SAFE if and only if its 95.45% upper-bound confidence slope (mean slope + 2*sigma_S) remains strictly within the ISRO threshold theta_safe <= 10.0 deg, and contains zero detected boulders or craters exceeding threshold dimensions:

ISRO 2-SIGMA SAFETY DECISION RULE

```
S_upper = S_cell_mean + 2.0 * sigma_S_cell
Cell Safe Condition: S_upper <= 10.0° AND CraterCount == 0 AND BoulderCount == 0
```

SECTION 6 — BBOX INTERSECTION-OVER-UNION (IoU) VALIDATION

Ground truth feature validation uses greedy bipartite bounding box matching evaluated at IoU >= 0.50:

IoU MATCHING METRIC

```
IoU(B_det, B_gt) = Area(B_det ") B_gt ) / Area(B_det "* B_gt )
```

SECTION 7 — 5-STAGE SHA-256 DETERMINISTIC EVIDENCE CHAIN

Stage 1: Input Buffer Hash	sha256(raw_bytes) — Seals raw telemetry buffer directly upon ingestion.
Stage 2: Calibrated Slope Hash	sha256(Float32Array(slope_map)) — Seals full-precision IEEE 754 slope matrix.
Stage 3: ISRO 24m Grid Hash	sha256(grid_classification_cells) — Seals discretized safety assignments.
Stage 4: Candidate Zones Hash	sha256(contiguous_landing_polygons) — Seals final touchdown zones.
Stage 5: Session Fingerprint	sha256(scan_0 || scan_1 || ... || scan_N) — Continuous immutable session log.

SECTION 8 — AES-256-GCM ENCRYPTION & ED25519 SIGNATURES

To guarantee end-to-end security and tamper resistance, all persisted flight logs are encrypted via AES-256-GCM using PBKDF2 key derivation (100,000 iterations, 32-byte salt, 12-byte initialization vector, 128-bit authentication tag). Mission dossiers and report summaries are signed using asymmetric Ed25519 elliptic-curve keypairs for non-repudiation.